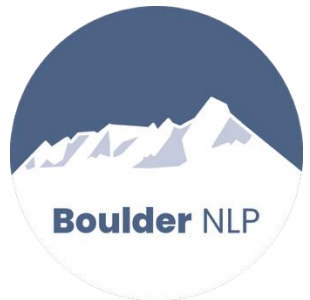


Natural Language Processing

Jim Martin -- Lecture 5
CSCI 5832



Today

- Wrap up evaluation of LMs.
- Classification and Logistic Regression

Entropy and Perplexity

$$H(T) = -\frac{1}{N} \log_2 P_m(T)$$

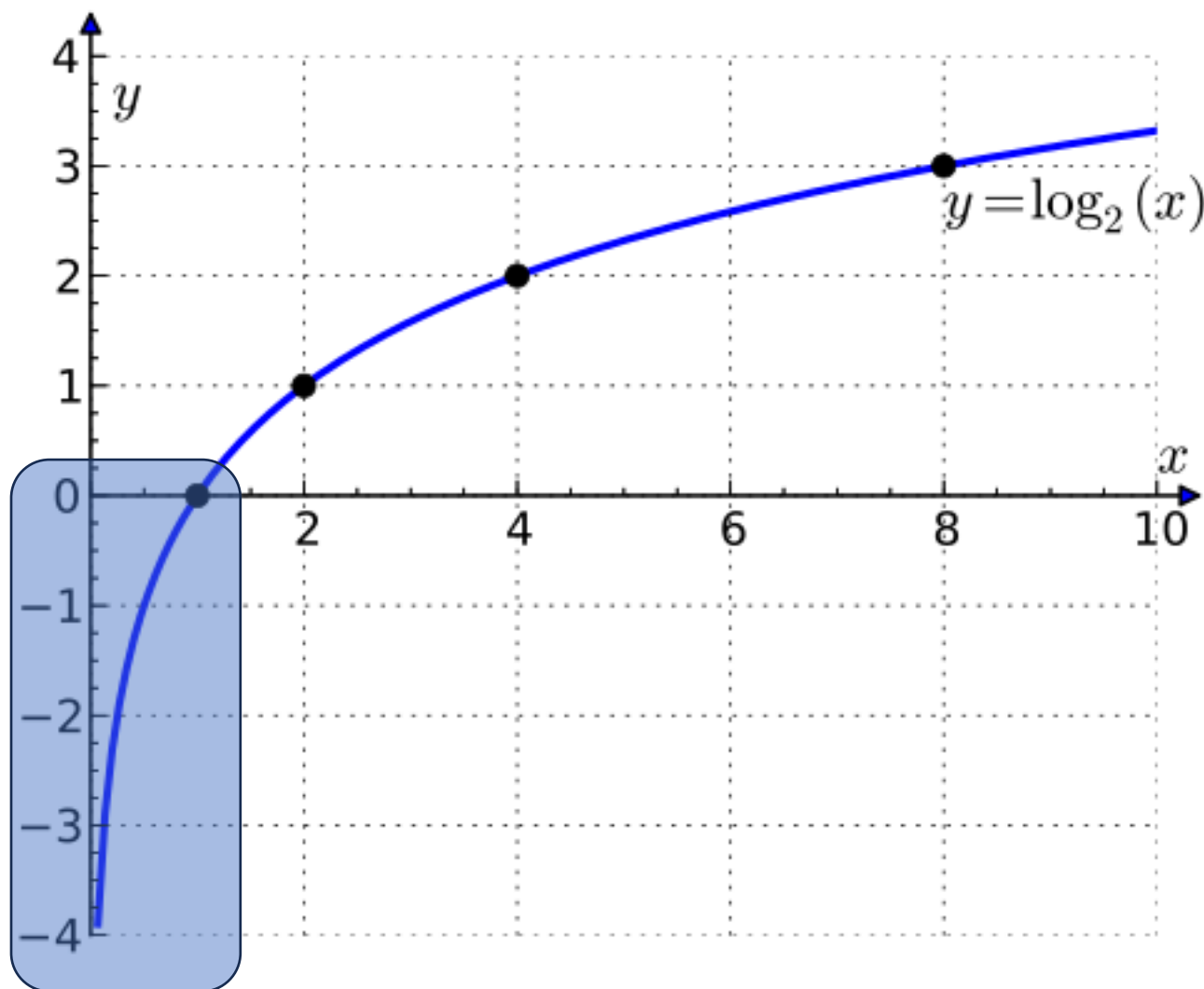
$$PP(T) = 2^{H(T)}$$

$$= -\frac{1}{N} \log_2 \prod_{i=1}^N P_m(w_i | w_{1:i-1})$$

$$H(T) = -\frac{1}{N} \sum_{i=1}^N \log P(w_i | w_{1:i-1})$$

Log of a probability

$$H(T) = -\frac{1}{N} \sum_{i=1}^N \log P(w_i | w_{1:i-1})$$



Perplexity and Entropy

$$PP(T) = 2^{H(T)}$$

Entropy		
Unigram	Bigram	Trigram
9.91	7.41	6.77

Perplexity		
Unigram	Bigram	Trigram
962	170	109

SemEval 2027

<https://semeval.github.io/SemEval2027/tasks>

Moving on (Chapter 4)

Objectives

- Classification as a task
 - Sentiment analysis as an example
- Gradient-based learning
 - Logistic regression as the example

Classification

- Classification is the task of examining an experience, event or entity and assigning it to a known class, or category.
- Classification, or categorization, is a central element to cognition. Humans perform this task effortlessly and without conscious awareness.
- By placing new instances into known categories, we can reason about instances based on what we know (or think we know) about members of that class.

Machine Classifiers

- Speech recognition
- Image classification
- Radiology analysis
- Robotic navigation
- Intrusion detection



Text Classification

The thing being classified is a piece of text... Some span of tokens for which we need to take some action.

An action that depends on its membership in some given class.



Spam detection



Sentiment analysis



Author identification



Deception detection



Language identification



Resume screening/routing

Sentiment Analysis

Extracting the sentiment that a writer has expressed towards some kind of entity in a text

- Movies, restaurants, hotels, consumer products, etc. Often expressed on commercial platforms (rotten tomatoes, yelp, tripadvisor, amazon,...).
- Or political orientation towards issues or candidates.
- As a categorical decision: positive, negative, neutral
- Or a numerical rating: star ratings for hotels, or % positive as on movie reviews



10/10 Exceptional

Feb 8, 2024

I love this classic hotel. She feels like a perfectly kept time warp. I prefer the original rooms in the old building but the newer wings rooms are more spacious and very nice too. You can walk all of downtown Boulder from the hotel. Great lobby and restaurants and bar.

6/10 Good

Feb 18, 2024

Must have been placed in the least desirable room on the ground level near the event/ conference

4/10 Fair

Jan 16, 2024

The room was so loud - it was my worst night of sleep at a hotel ever. I'm not sure why they would put any guest in this room. It's the first room after any guest crosses the bridge from the historic hotel to the annex with a public restroom across the hall. We heard every conversation as people went back to their rooms after their night out and every conversation first thing in the morning. I commented that it was my worst night of sleep in a hotel when we checked out and the receptionist said he would let management know to see what they could do to compensate us. I never heard anything back. So disappointing.

ed at 1:30 am, we morning. It's also customer service to the hotel are

Spam Detection

To: James Martin



This message is from a mailing list.

Unsubscribe



A. Vitolin
Reply

Reply-To: bent@uncurved.org

Dear sir/madam,

Kindly confirm If you can be trusted to run a humanitarian charity project sponsored by an incentive/compensation to you for your trust in me. If interested, reply me for further details.

Thanks & regards
Arachi Vitolin.

This message was sent automatically because your Sirius XM subscription has expired.

sxm

Your Sirius XM membership has expired!

Unfortunately, we were unable to renew your subscription. If you choose to keep your subscription active, you should update your profile.

As part of our loyalty program, you can now extend for 90 days for FREE!

RENEW SUBSCRIPTION NOW

*After signing up, you have to insert your credit card details for validation of your account. We **will not** withdraw any amount.

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James Martin
artin

Details

It is very important that you read this email

★Check If You have been successfully approved★

Please Check Your account

Congratulations! james.h.martin✓

! Could Get \$10 Billion In Roundup Lawsuit

Please confirm your INFORMATION NOW!

YOUR ACCOUNT INFORMATION :

Name: james.h.martin [Verify*](#)

EMAIL: james.h.martin@gmail.com [Verify*](#)

Date: Tue, 13 Feb 2024 15:46:51 +0000 [Verify*](#)

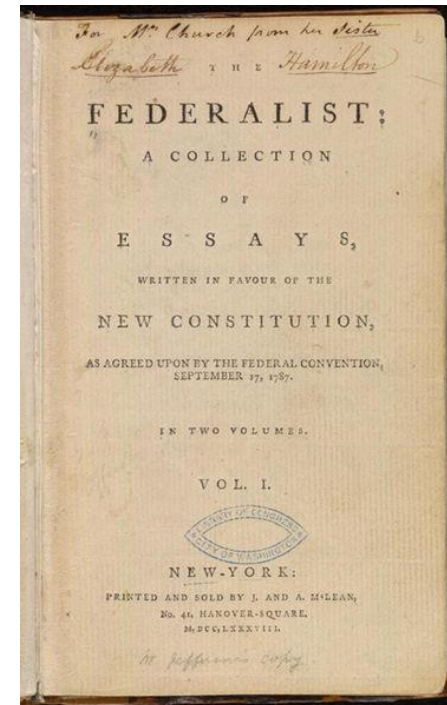
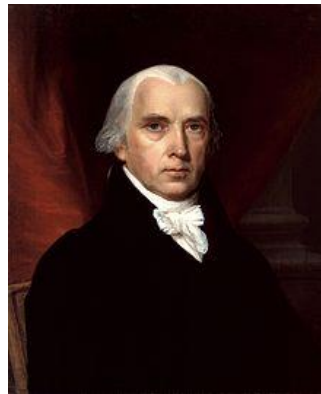
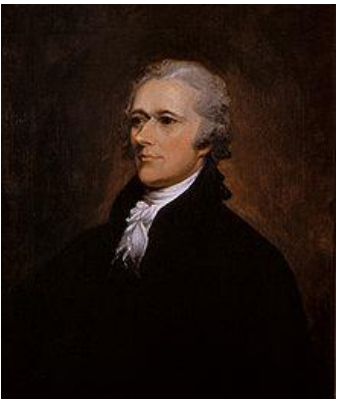
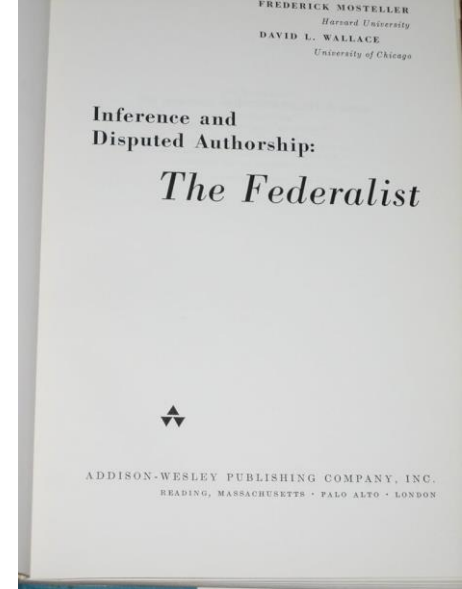
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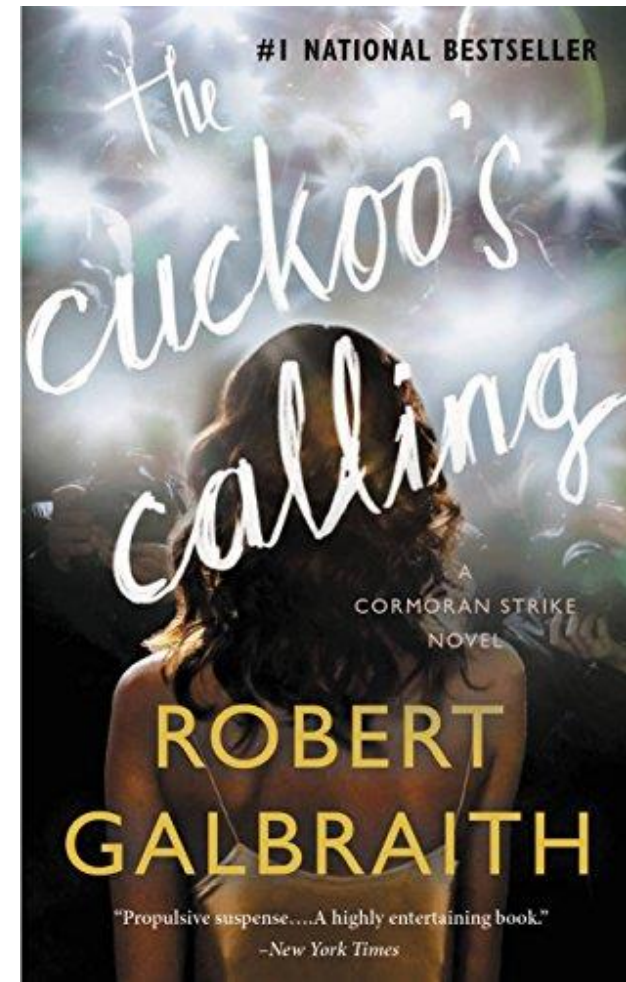
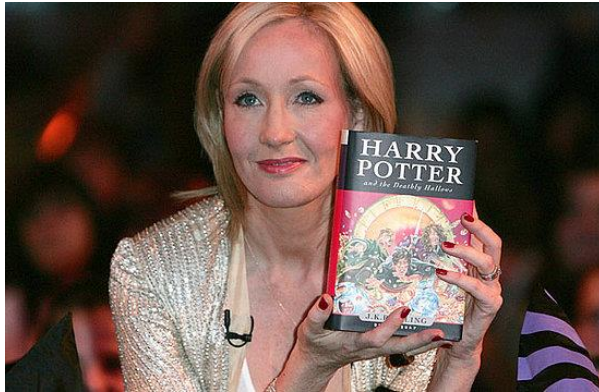
Author Identification

- 1787-8: Anonymous essays written to convince New York to ratify new U.S. Constitution: John Jay, James Madison, Alexander Hamilton.
- Authorship of some of the letters in dispute



More Contemporary Examples

- *The Cuckoo's Calling* by “Robert Galbraith”



More Examples

■ *NY Times “Anonymous” editorial*

The New York Times

I Am Part of the Resistance Inside the Trump Administration

I work for the president but like-minded colleagues and I have vowed to thwart parts of his agenda and his worst inclinations.

Sept. 5, 2018

The Times is taking the rare step of publishing an anonymous Op-Ed essay. We have done so at the request of the author, a senior official in the Trump administration whose identity is known to us and whose job would be jeopardized by its disclosure. We believe publishing this essay anonymously is the only way to deliver an important perspective to our readers. We invite you to submit a question about the essay or our vetting process here. [Update: Our answers to some of those questions are published here.]

President Trump is facing a test to his presidency unlike any faced by a modern American leader.

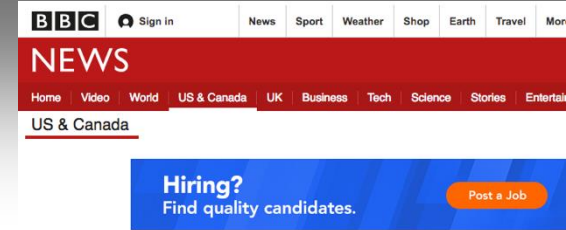
It's not just that the special counsel looms large. Or that the country is bitterly divided over Mr. Trump's leadership. Or even that his party might well lose the House to an opposition hellbent on his downfall.

Here's one big clue to the identity anonymous op-ed writer

Analysis by Chris Cillizza, CNN Editor-at-large
Updated 7:42 PM ET, Thu September 6, 2018



(CNN) — By the time you read this, practically every person who works in the Trump administration — or even resides within the Trump administration — will have started



NYT Trump column: Linguistic clues to White House insider?

By Roland Hughes
BBC News

6 September 2018

Share



Sherlock would no doubt approve
We all have our own distinctive style of writing and speaking. Trying to hide those quirks is like trying to repress a part of our character.

This style is what can help you identify an author from reading only one paragraph of their work. But what happens if the author doesn't want to be identified?

POLITICS

The Obvious Suspect

The quest to unmask the New York Times op-ed writer has been filled with speculation. But the article's prose points to one person in particular.

By WILLIAM SALETAN

SEPT 07, 2018 • 1:15 PM



Jon Huntsman visits Gerson Lehrman Group on Dec. 2, 2014.
Photo by Andrew Tobin/Getty Images for GLG

Who wrote the *anonymous op-ed* against President Trump in Wednesday's New York Times? All we know for certain is what the Times disclosed: that it's a "senior official in the Trump administration." But the most likely author, based on the op-ed's content and style, is the U.S. ambassador to Russia, Jon Huntsman.

Text Classification

Formally, we'll be working with the following setup...

- We have an input text drawn from some set of texts $t \in \mathbb{T}$
- And we have a fixed set of classes $\mathbb{C} = \{c_1, c_2, c_3, \dots, c_n\}$
- We want as output a single predicted class selected $c \in \mathbb{C}$

Machine Learning Approaches

- We have available a set of tuples consisting of a representative set of texts with their correct labels.
- From those examples, we'd like to learn a classifier that maps texts to their corresponding classes

$$\{(t_i, c_i)\}^n$$

$$\theta : \mathbb{T} \rightarrow \mathbb{C}$$

Probabilities

- But instead of a discrete classification we will want a probabilistic decision instead. That is, what is the probability that this text belongs to this class?

$$P(c|t)$$

- For example, the probability that a given email is spam.
- And when asked to give a definitive answer we might reply with the class with the highest probability.

$$c = \operatorname{argmax}_{c \in C} P(c|t)$$

Logistic Regression

- Logistic regression models estimate the conditional probability of a class assignment given a text:
 - A scoring function that makes use of
 - Numerical representation of the text
 - Numerical weights associated with those representations
 - A supervised scheme that allows us to learn the weights.

$$P(c|t)$$

Scoring

- Given features and weights, we can assign a score to each class for a given document using an affine function. An affine function is just a linear function with an offset, or intercept.

$$z = b + w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n$$

- For now, let's assume we're dealing with a binary classification problem. Whether a text belongs to a particular class, or not.
 - The score z then represents a document's score as a member of the positive class.

Scoring

$$z = b + w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n$$

$$z = \left(\sum_{i=1}^n w_i x_i \right) + b$$

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

$$z = \mathbf{w}^\top \mathbf{x} + b$$

Scoring

To eliminate some notational complexity, we can incorporate the bias term as an additional weight and set the corresponding feature value to 1.

$$w_0 = b$$

$$x_0 = 1$$

$$z = w_0x_0 + w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n$$

$$z = \mathbf{w}^T \mathbf{x}$$

Scoring

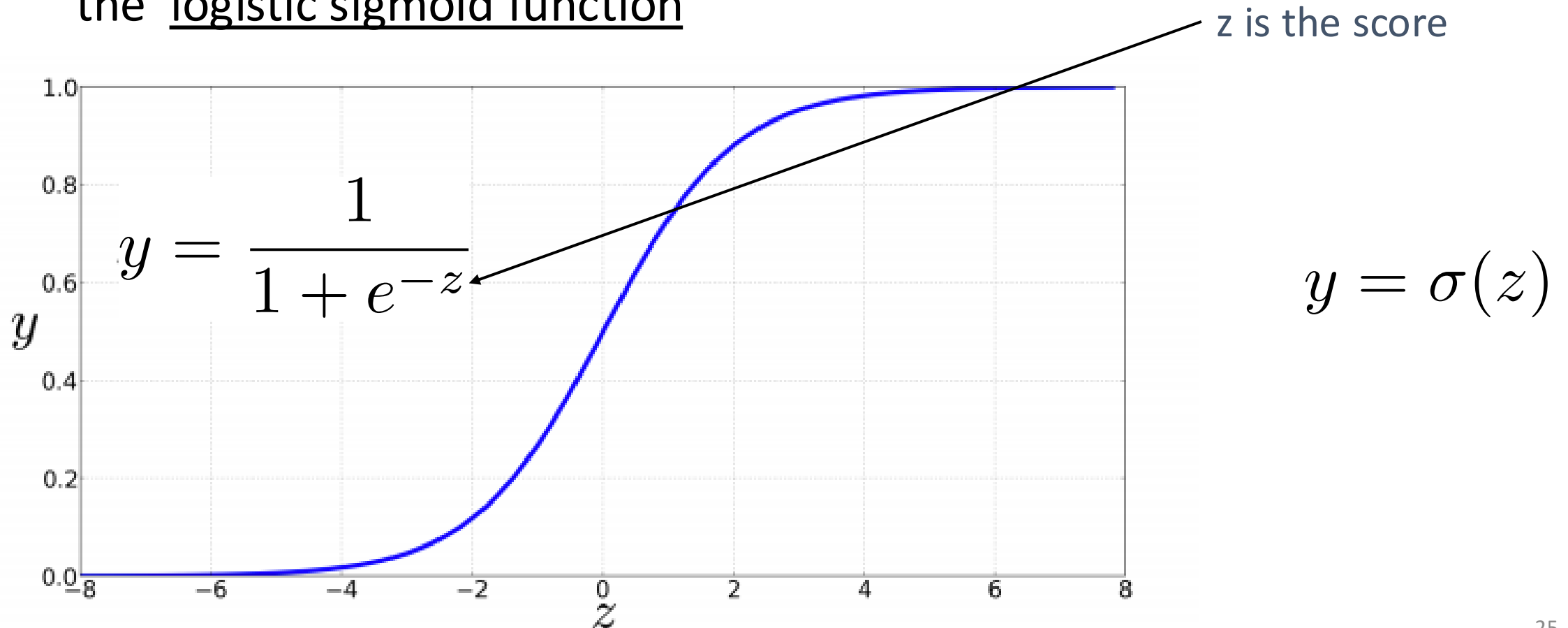
- The real power of logistic regression comes from its conversion of that the score into a probability

$$P(c|t)$$

- This will allow us to handle uncertainty in our classifications
 - Useful in downstream applications
 - And, it provides us with a learning scheme

Scoring

- To accomplish this, we'll squash the scores to between 0 and 1 using the logistic sigmoid function

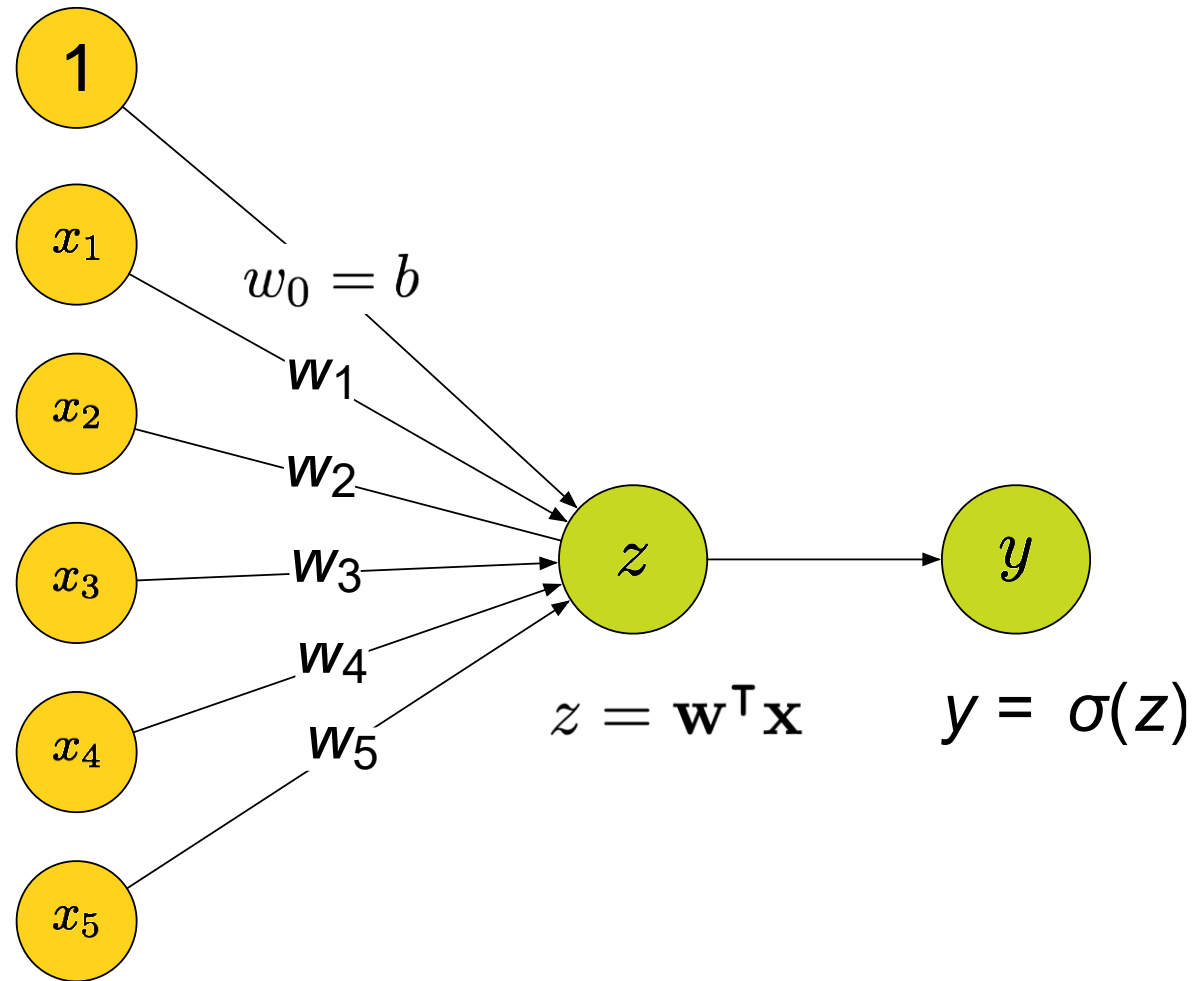


Scoring

$$\begin{aligned} P(y = 1|x) &= \sigma(\mathbf{w}^\top \mathbf{x}) \\ &= \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}} \end{aligned}$$

$$P(y = 0|x) = 1 - \sigma(\mathbf{w}^\top \mathbf{x})$$

Scoring as a Network Model



Representations (Features)

With this approach we're not classifying texts directly, we're classifying texts based on their representations.

- Here, we'll represent texts as sets of discrete features. Logistic regression uses these features as clues as to the correct class. In general, the features we will employ have the following characteristics:
 1. They can be reliably and efficiently extracted
 2. They are hand crafted based on domain knowledge and data analysis
 3. And there may be a whole lot of them

Sentiment Analysis: Hotel Review

GORGEIOUS HOTEL! I stayed here for my honeymoon. Charleston is so fabulous, and this is totally the hotel to stay at. I'm so glad I saved up my money for it. I've only ever stayed at gross cheap hotels. This one had nice soft bed sheets, and a tasty breakfast, and a garden patio...it was hard to make myself go out and explore the city! They even put little cards in the room to say what the weather was going to be like the next day. So charming.

Common Sentiment Features

- Lists of positive and negative words
 - Called a "sentiment lexicon". Typically, a combination of generic and domain specific words. Represented as counts of either tokens or types
- The presence of "no" (and other polarity expressions) in the review
- Use of personal pronouns (I, we, they, etc.)
- Presence of specific kinds of punctuation (like !!!)
- Review length
- Case patterns
 - Lower case, upper case, title case, camel-case, etc.

Sentiment

GORGEIOUS HOTEL! I stayed here for my honeymoon. Charleston is so **fabulous**, and this is totally the hotel to stay at. I'm so glad I saved up my money for it. I've only ever stayed at gross cheap hotels. This one had **nice** **soft** bed sheets, and a **tasty** breakfast, and a garden patio...it was hard to make myself go out and explore the city! They even put little cards in the room to say what the weather was going to be like the next day. So **charming**.

Sentiment

GORGEIOUS HOTEL! I stayed here for my honeymoon. Charleston is so **fabulous**, and this is totally the hotel to stay at. I'm so glad I saved up my money for it. I've only ever stayed at **gross** **cheap** hotels. This one had **nice** **soft** bed sheets, and a **tasty** breakfast, and a garden patio...it was hard to make myself go out and explore the city! They even put little cards in the room to say what the weather was going to be like the next day. So **charming**.

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Sentiment

GORGEOUS HOTEL! I stayed here for my honeymoon. Charleston is so fabulous, and this is totally the hotel to stay at. I'm so glad I saved up my money for it. I've only ever stayed at gross cheap hotels. This one had nice soft bed sheets, and a tasty breakfast, and a garden patio...it was hard to make myself go out and explore the city! They even put little cards in the room to say what the weather was going to be like the next day. So charming.

Representing Feature Values

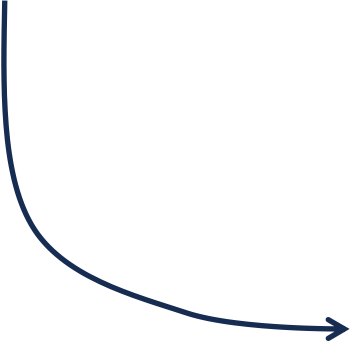
For our scoring function to work, features need to have numerical values

- Some features are naturally captured as numbers
 - Lengths, counts, etc.
- Nominal or categorical features can be captured as binary values

Sentiment Example

GORGEOUS HOTEL! I stayed here for my honeymoon.
Charleston is so **fabulous**, and this is totally the hotel to stay at.
I'm so glad I saved up my money for it. I've only ever stayed at
gross cheap hotels. This one had **nice soft** bed sheets, and a
tasty breakfast, and a garden patio...it was hard to make myself
go out and explore the city! They even put little cards in the
room to say what the weather was going to be like the next
day. So **charming**.

1. 6 positive words
2. 2 negative words
3. 1 exclamation point
4. 2 uses of all CAPs
5. 7 uses of personal pronouns
6. Log of token count = 4.49
7. Log of type count = 4.23



[6, 2, 1, 2, 7, 4.49, 4.23]

Scoring

GORGEOUS HOTEL! I stayed here for my honeymoon. Charleston is so fabulous, and this is totally the hotel to stay at. I'm so glad I saved up my money for it. I've only ever stayed at gross cheap hotels. This one had nice soft bed sheets, and a tasty breakfast, and a garden patio...it was hard to make myself go out and explore the city! They even put little cards in the room to say what the weather was going to be like the next day. So charming.



[6, 2, 1, 2, 7, 4.49, 4.23]

Now, let's assume we already have weights for each of these features along with a bias term of 0.1.

[0.1 , -0.0848, -0.0983, 0.1056 , -0.0870, 0.0039 , -0.0833, 0.0502]

Scoring

GORGEOUS HOTEL! I stayed here for my honeymoon. Charleston is so **fabulous**, and this is totally the hotel to stay at. I'm so glad I saved up my money for it. I've only ever stayed at **gross cheap** hotels. This one had **nice soft** bed sheets, and a **tasty** breakfast, and a garden patio...it was hard to make **myself** go out and explore the city! They even put little cards in the room to say what the weather was going to be like the next day. So **charming**.

$$\mathbf{w} = [0.1, -0.0848, -0.0983, 0.1056, -0.0870, 0.0039, -0.0833, 0.0502]$$

$$\mathbf{x} = [1, 6, 2, 1, 2, 7, 4.49, 4.23]$$

$$\begin{aligned} P(y = 1|\mathbf{x}) &= \sigma(\mathbf{w}^\top \mathbf{x}) \\ &= \sigma([0.1, -0.0848, -0.0983, 0.1056, -0.0870, 0.0039, -0.0833, 0.0502] \cdot [1, 6, 2, 1, 2, 7, 4.49, 4.23]) \\ &= \sigma(-0.8079) \\ &= 0.3083 \end{aligned}$$

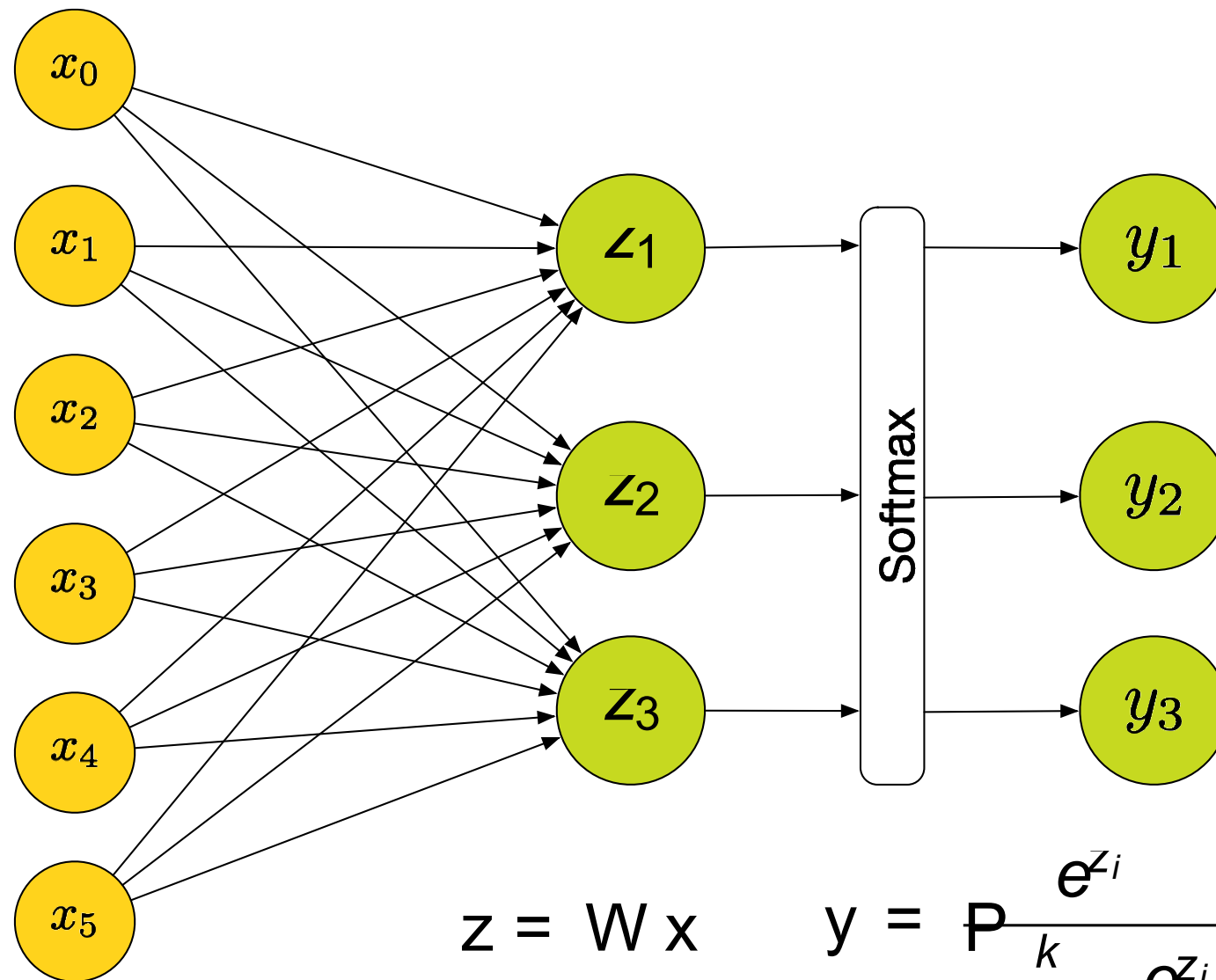
Multiclass Models

- Most practical applications have more than two classes. Even simple examples like spam detection and sentiment generally use three classes:
 - Spam, not spam, and 🧑
 - Positive, negative and 🧑
 - Hamilton, Madison, Jay
 - Document categorization may have thousands of possible classes.
- So, we need to adjust our scoring function!!

Multiclass Models

- The easiest way to think about the multiclass case is as a set of binary classifications. where Each class has its own set of feature weights to assess the score for its class.
- The resulting scores can then be normalized to produce a probability distribution over all the classes.
 - Softmax
- Note by multiclass, we mean choosing a single class from a set of classes. NOT the same as assigning multiple labels to a text. That's a different problem!

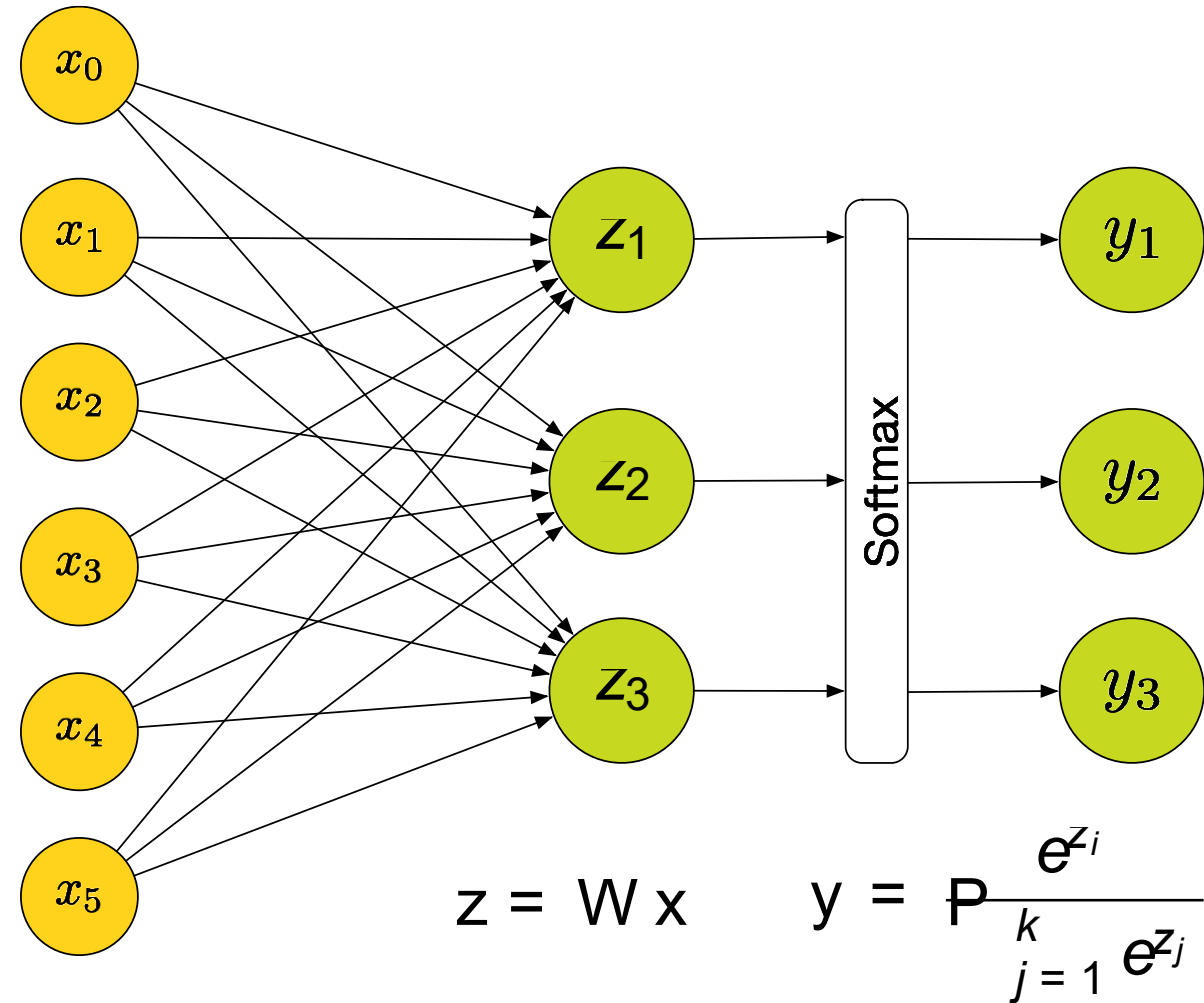
Softmax



$$z = W x \quad y = \text{P} \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

Softmax

- We maintain the same set of features as in the binary setting.
- But now each class has its own set of weights for those features.
- We can capture the weights as a $k \times n$ matrix, where k is the number of classes and n is the number of features.
- Multiplying that matrix by the input vector gives us a vector of scores (one for each class).



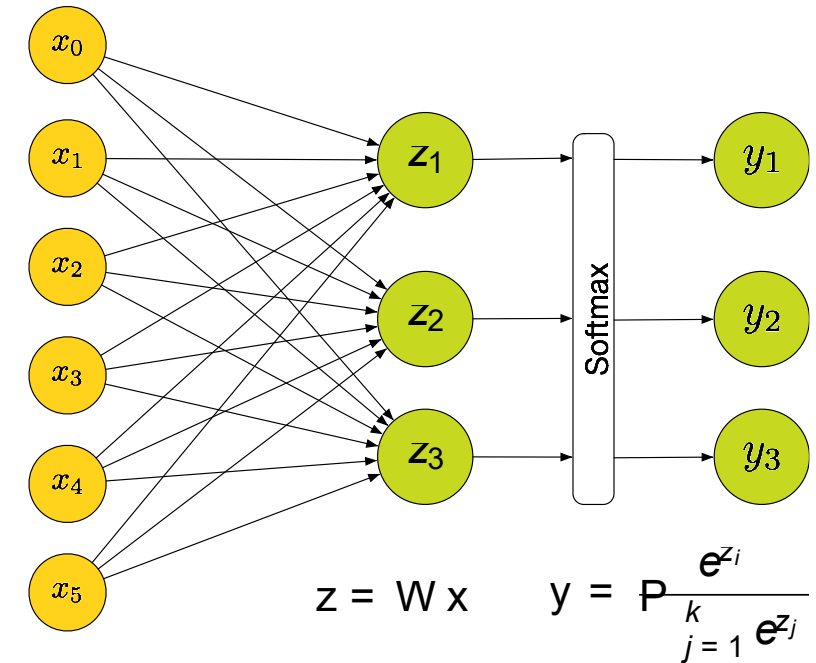
Softmax

The raw score for a single class c .

$$P(y = c|x; \mathbf{w}) = \frac{e^{z_c}}{\sum_{j=1}^k e^{z_j}}$$

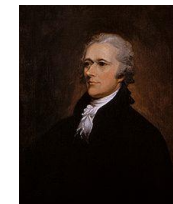
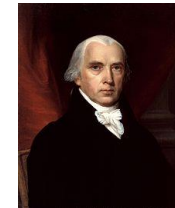
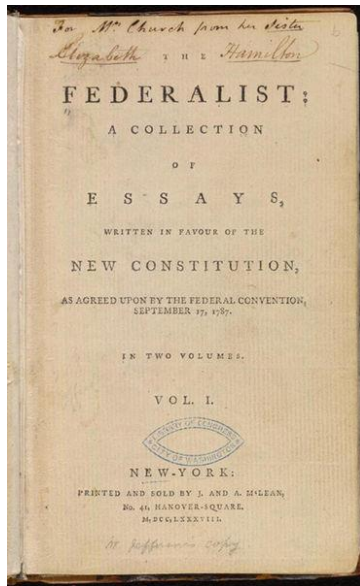
The probability that a text x belongs to class c .

This sum of scores across all classes.

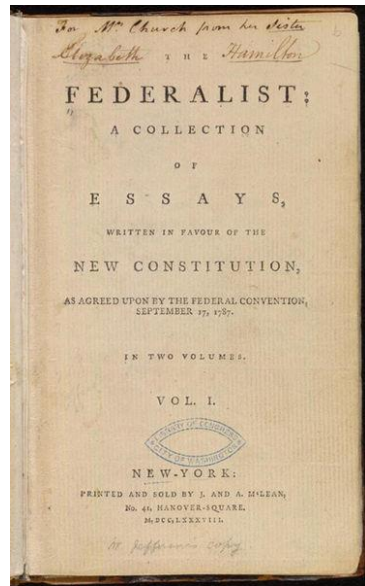


$$z_c = \mathbf{W}_c : \mathbf{x}$$

Back to the Federalists



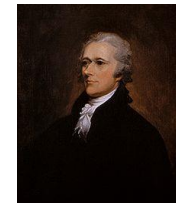
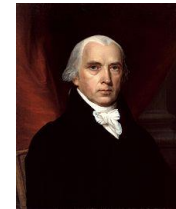
Back to the Federalists



1. Extract features

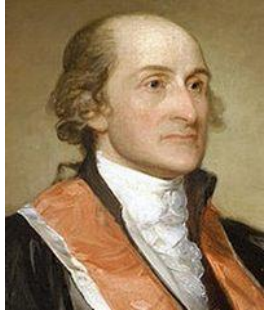
2. Compute z scores

3. Compute
Softmax



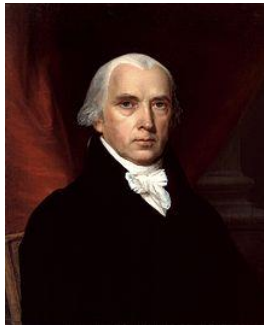
Federalist Example

$$P(y = c|x; \mathbf{w}) = \frac{e^{z_c}}{\sum_{j=1}^k e^{z_j}}$$



$$e^{z_{\text{jay}}} = 0.00039$$

$$\begin{aligned} P(\text{Jay} \mid \text{text}) &= .00039 / (.00039 + .0409 + .0000017) \\ &= .00946 \end{aligned}$$



$$e^{z_{\text{madison}}} = 0.0409$$

$$\begin{aligned} P(\text{Madison} \mid \text{text}) &= .0409 / (.00039 + .0409 + .0000017) \\ &= .98959 \end{aligned}$$



$$e^{z_{\text{hamilton}}} = 0.0000017$$

$$\begin{aligned} P(\text{Hamilton} \mid \text{text}) &= .0000017 / (.00039 + .0409 + .0000017) \\ &= .00004 \end{aligned}$$

Recap

- We're using a linear function to produce scores that reflect how well a given text fits a given category:
- In the binary case we have proposed to turn those scores into probabilities using the logistic sigmoid function:
- But why is this a sensible thing to do? Why the sigmoid or softmax vs. some other that might turn arbitrary scores into probabilities?

$$z = \mathbf{w}^T \mathbf{x}$$

$$p = \frac{1}{1 + e^{-z}}$$

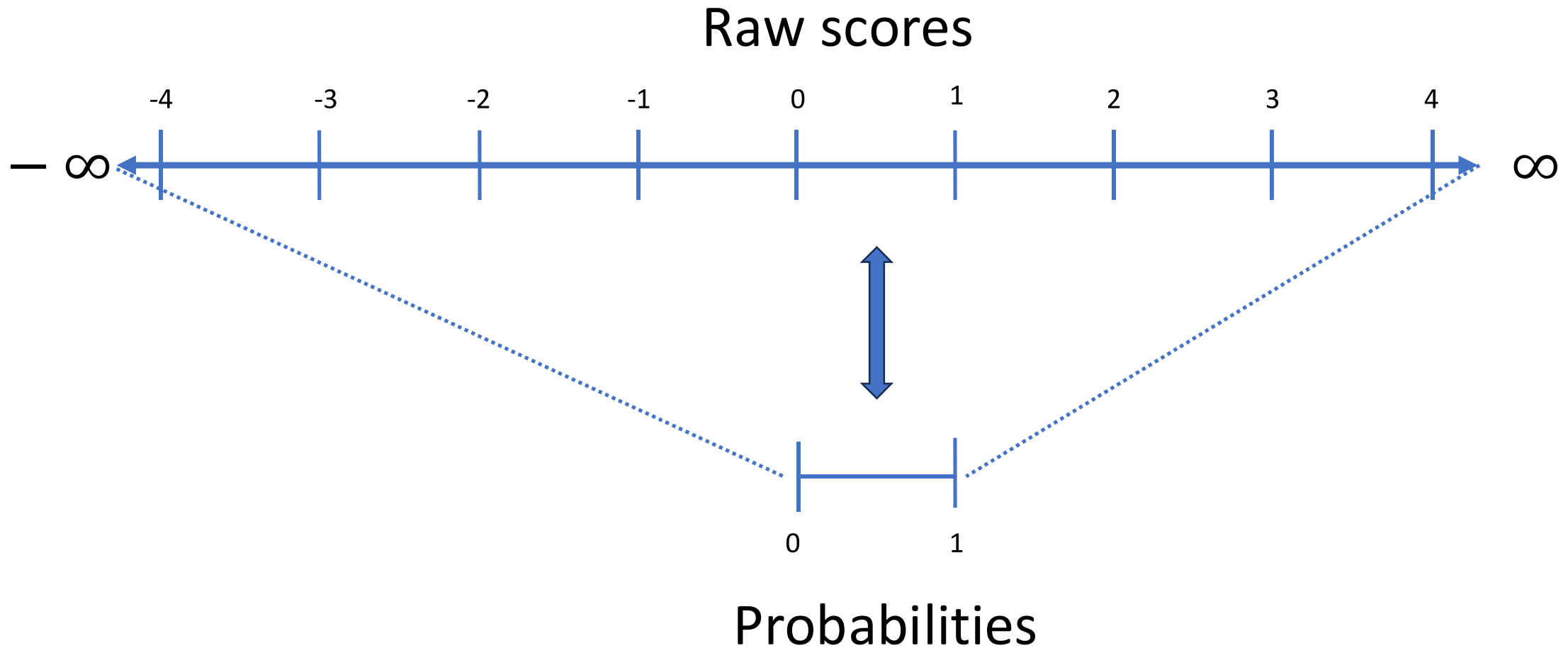
Motivation

Before diving in to learning, let's review some details with our scoring function.

$$z = \mathbf{w}^T \mathbf{x}$$

- We haven't restricted our weights, the bias term, or the feature values to anything other than real numbers.
- Therefore, the possible range of scores (z) can extend from $-\infty$ to ∞ . That is, the possible scores lie anywhere along the real number line.

To Infinity and Beyond



Step 1: From Probabilities to Odds

Without any loss of information, in the binary case, we can express probabilities as simple odds: the ratio of one possibility to the other.

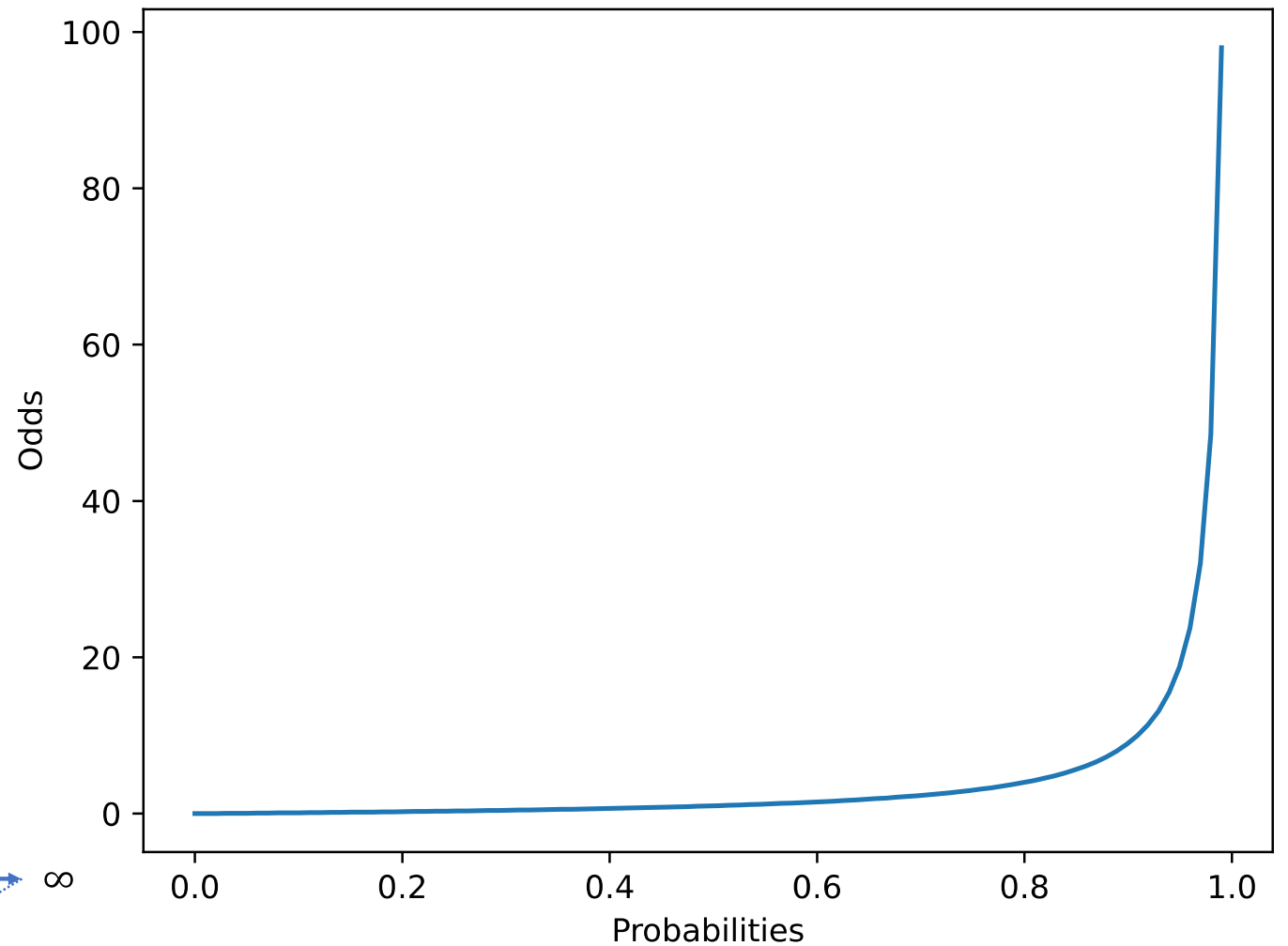
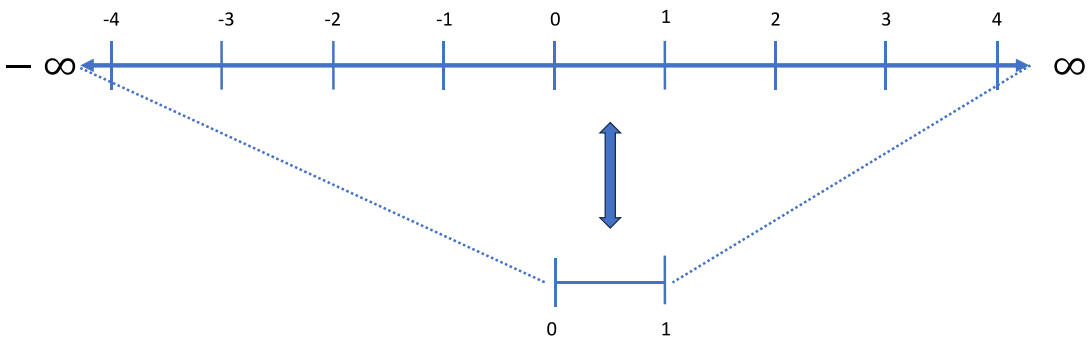
.95	19 to 1	19
.8	4 to 1	4
.75	3 to 1	3
.5	1 to 1	1
.25	1 to 3	.3333
.2	1 to 4	.25
.05	1 to 19	.0526

$$\frac{p}{1 - p}$$

For example, if the probability of some (binary) event is 0.75 then the odds are $0.75/0.25$ or 3/1

Step 1: Probabilities to Odds

While probabilities
range from 0 to 1, odds
range from 0 to infinity.



Step 1: Probabilities to Odds

- Moving on we can observe that there's an annoying asymmetry to our odds values.
- 19 to 1 and 1 to 19 really should be related somehow, but they're not obviously --- we have a 19 and a .0526.

<u>.95</u>	<u>19 to 1</u>	<u>19</u>
.8	4 to 1	4
.75	3 to 1	3
.5	1 to 1	1
.25	1 to 3	0.3333
.2	1 to 4	0.25
<u>.05</u>	<u>1 to 19</u>	<u>0.0526</u>

Step 2: Odds to Log Odds

- Let's take the log of the odds!

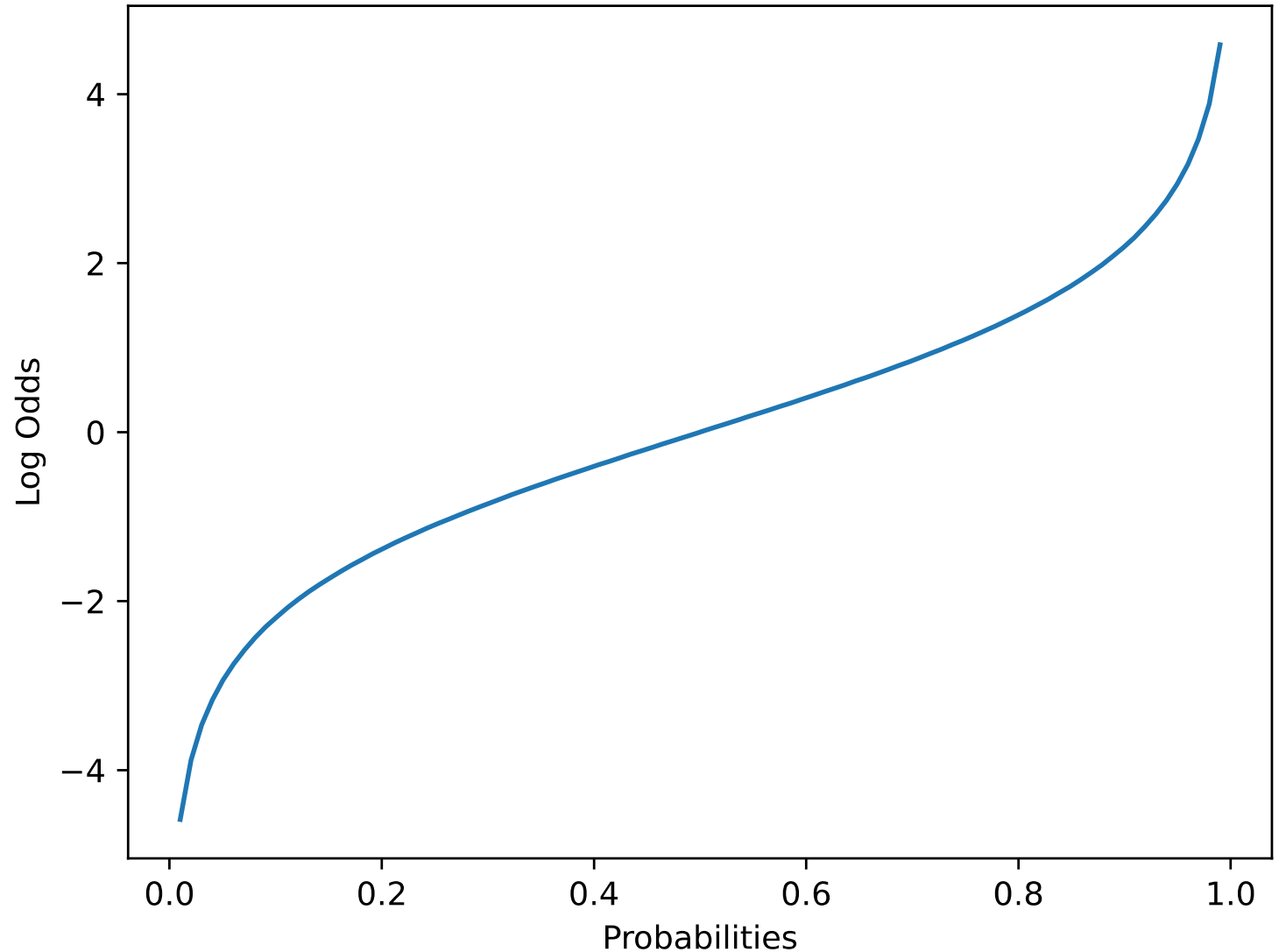
$$\ln \left(\frac{p}{1-p} \right)$$

.95	19 to 1	19	<u>2.944</u>
.8	4 to 1	4	1.38
.75	3 to 1	3	1.098
.5	1 to 1	1	0
.25	1 to 3	.3333	-1.098
.2	1 to 4	.25	-1.38
.05	1 to 19	.0526	<u>-2.944</u>

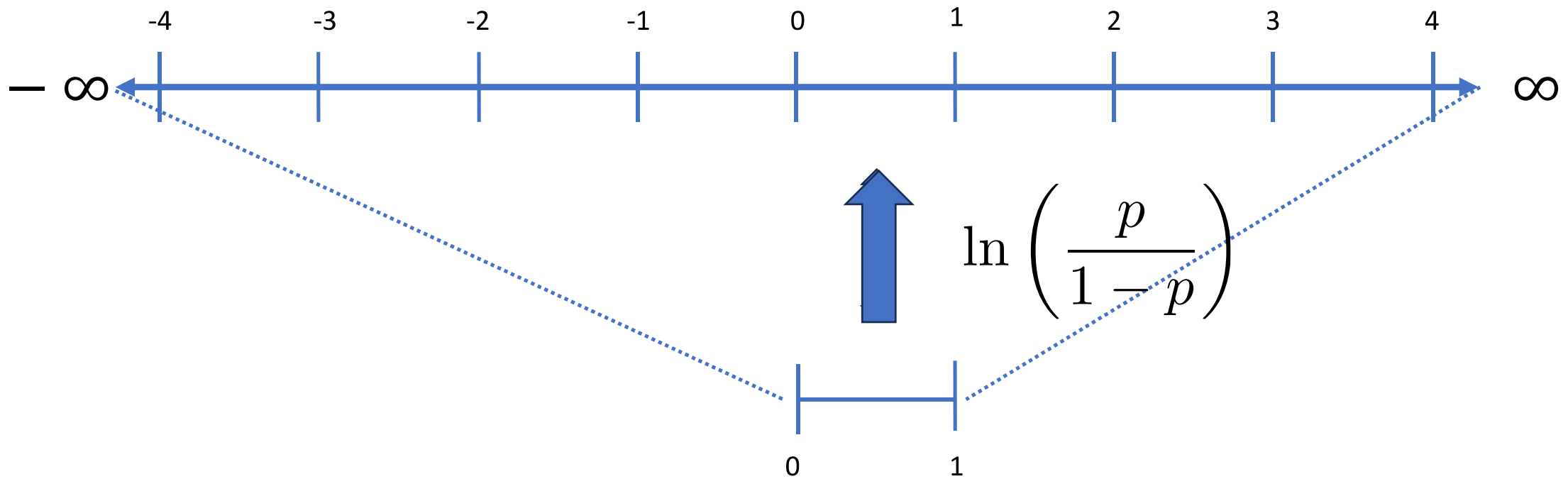
- Symmetry!
- And we've found a way to map probabilities to the real number line. Centered around 0 (no information to distinguish).

Step 2: Odds to Log Odds

- While probabilities range from 0 to 1, log odds range from $-\infty$ to ∞ .



Reals to Probabilities



Logit

- The log of the odds of the probability of an event is called the logit.

$$\text{logit} = \ln \left(\frac{p}{1 - p} \right)$$

- The scores z from our $(w^T x)$ computation are logits. During learning we will be searching for weights w that produce the right score logits.
- So, now we know how take a probability and produce its logit. How do we go the other way? Given a logit, produce a probability.
- We've already seen the answer, but let's do some algebra to be sure.

$$z = \ln \left(\frac{p}{1-p} \right) \quad \text{Let's call our logit } z.$$

$$e^z = \frac{p}{1-p} \quad \text{Exponentiate both sides to rid us of that pesky log.}$$

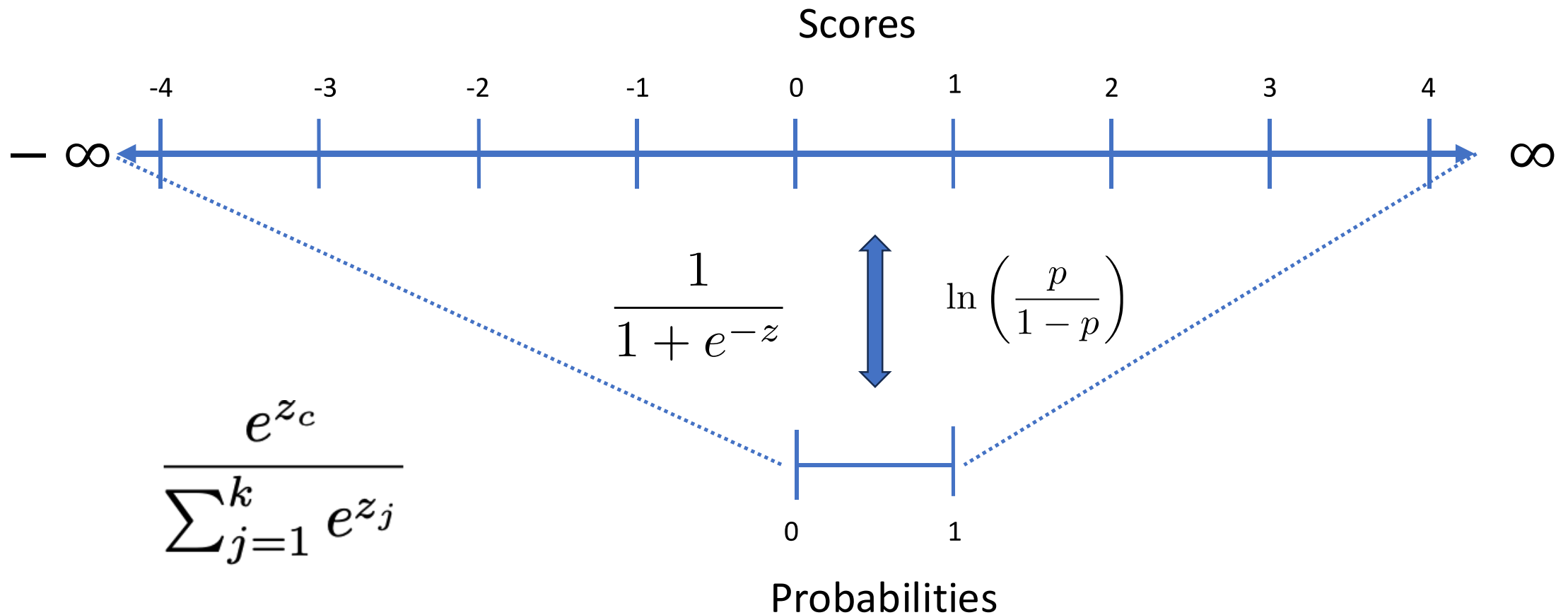
$$e^z(1-p) = p \quad \text{Some algebra to isolate } p.$$

$$e^z - pe^z = p$$

$$e^z = p + pe^z \quad \text{And voila! The sigmoid – } p \text{ in terms of } z.$$

$$e^z = p(1 + e^z) \quad p = \frac{e^z}{1 + e^z} = \frac{1}{1 + \frac{1}{e^z}} = \frac{1}{1 + e^{-z}}$$

To Infinity and Beyond



Take Aways

- The logit is the log of the odds of some event. Where the odds is the ratio of the probability of an event being the case to the probability of it not being the case.
- The raw scores from our linear classifier are logits. Logits will play a central role later in the course as we move from simple models to deep learning.
- Our next big task is to come up with a learning scheme to acquire the proper weights w , so that our scores represent logits appropriate to our task.